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Investigation on the Air Supply Subsystem of PEMFC Based on Sliding Mode Control and Model Predictive Control

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Abstract:

To improve the performance and stability of proton exchange membrane fuel cell (PEMFC), this paper proposes a hybrid control strategy that integrates sliding mode control (SMC) with model predictive control (MPC). Based on a control framework that combines global optimization with local compensation, the proposed method optimizes PEMFC operating conditions while improving system robustness. A PEMFC air supply subsystem model was developed in Simulink to conduct numerical simulations of SMC, MPC, and the hybrid control strategy. The results show that the hybrid approach achieves the smallest overshoot, reduced by 60% and 28% compared with SMC and MPC, respectively. It also attains the shortest settling time, decreased by 20% and 52%, and the fastest disturbance recovery time, improved by 50% and 37.5%. Overall, the hybrid strategy improved dynamic metrics for oxygen excess ratio, voltage stability, and net power under tested load steps.

Keywords: PEMFC; Sliding mode control; model predictive control; Hybrid control; Air supply subsystem

Nomenclature

$PEMFC$	Proton Exchange Membrane Fuel Cell
SMC	Sliding Mode Control
MPC	Model Predictive Control
J_{cp}	Rotational inertia of the air compressor
T_{cm}	Motor torque
T_{cp}	Load torque
η_{cm}	Mechanical efficiency of the motor
V_{cm}	Air compressor voltage
C_p	Specific heat capacity of air at constant pressure
P_{atm}	Atmospheric pressure
P_{sm}	Supply line pressure
η_{cp}	Air compressor efficiency
T_{atm}	Atmospheric temperature
W_{cp}	Air compressor flow rate
V_{sm}	Supply pipe volume
T_{cp}	Air compressor outlet temperature
$K_{ca,in}$	Cathode outlet orifice constant
$W_{ca,in}$	Total flow rate entering the cathode
Ω_{atm}	Humidity ratio at standard atmospheric pressure
M_a	Molar mass of air
ϕ_{atm}	Ambient relative humidity
n	Number of individual cells comprising the stack

I	Stack current
V_{ca}	Cathode volume
T_{st}	Stack temperature

1. Introduction

In recent decades, global energy demand has continued to increase, with traditional fossil fuels such as coal, oil, and natural gas still dominating the global energy supply structure. However, the extensive consumption of fossil fuels has led to severe environmental and ecological challenges, particularly the large-scale emissions of carbon dioxide and methane, which are recognized as major drivers of global climate change. Consequently, the substitution of conventional fossil fuels with clean and sustainable energy sources has become one of the most critical strategies for reducing carbon emissions^[1].

Hydrogen is one of the most abundant elements on Earth, and its utilization as an energy carrier produces only water as a by-product, making it a promising solution for facilitating the global energy transition and mitigating climate change^[2,3]. Compared with traditional fossil fuel-based energy conversion technologies, fuel cells exhibit advantages such as high energy conversion efficiency and environmentally benign operation, which endow them with significant potential in applications including transportation, aerospace, and stationary power generation systems^[4,5]. Since the beginning of the 21st century, increasing environmental concerns and the growing demand for new energy vehicles have accelerated the development of fuel cell vehicles worldwide^[6,7,8]. Among the various types of fuel cells, proton exchange membrane fuel cells (PEMFCs) have attracted considerable attention owing to their low operating temperature, fast start-up characteristics, and high-power density, making them particularly suitable for automotive and portable power applications^[9].

Sliding mode control has played a pivotal role in the advancement of proton exchange membrane fuel cell systems, exemplifying the mutually reinforcing relationship between modern control theory and clean energy technologies. Compared with conventional control strategies, SMC has attracted considerable attention due to its inherent robustness against parameter variations and external disturbances, superior tracking performance, and enhanced system stability^[10,11]. Numerous studies have investigated the application of SMC in PEMFC-related systems. Laghrouche *et al.* applied sliding mode control to PEMFC gas supply systems and demonstrated its effectiveness in the presence of system uncertainties^[12,13]. Wang *et al.* proposed an adaptive sliding mode controller and validated its performance experimentally, showing that the controller achieves rapid dynamic response while effectively suppressing oscillations^[14]. Silaa *et al.* integrated an adaptive stochastic gradient descent method into a sliding mode control framework, with simulation results indicating a significant reduction in chattering while maintaining robustness^[15]. Abbaker *et al.* introduced a delay-based algorithm into sliding mode control, and simulation studies

revealed improved response speed and overall system performance in PEMFC applications^[16]. To further enhance control performance, Fang *et al.* developed an adaptive sliding mode controller based on fuzzy logic, with experimental results demonstrating notable improvements in the efficiency and reliability of PEMFC systems^[17]. Li *et al.* proposed a control scheme combining input-output linearization with sliding mode control, achieving effective regulation of the oxygen excess ratio and cathode pressure^[18]. Javaid *et al.* introduced a fast terminal integral sliding mode control (FTISMC) strategy for oxygen excess ratio regulation in PEMFC power systems, with results confirming the elimination of chattering and improved system efficiency^[19]. Matraji *et al.* employed a second-order sliding mode controller to regulate the oxygen excess ratio, demonstrating excellent transient response characteristics^[20]. Zhu *et al.* proposed an adaptive sliding mode controller validated on an experimental platform, further confirming the effectiveness of SMC-based approaches in PEMFC control^[21]. In addition, Derakhshannia *et al.* designed a predictive sliding mode control strategy and verified its robustness and effectiveness through simulation studies^[22].

The research history of model predictive control in PEMFC systems reflects the progressive integration of advanced control theory with clean energy technologies. As traditional control strategies proved insufficient for addressing strong multivariable coupling and the stringent performance requirements of high-power, highly dynamic PEMFC systems, MPC emerged as a promising alternative due to its inherent capability to handle multivariable interactions and operational constraints^[23,24]. A growing body of research has explored the application of MPC in various PEMFC subsystems. Zhang *et al.* introduced MPC into the thermal management system of a PEMFC and compared its performance with that of a proportional-integral (PI) controller, demonstrating that MPC enables effective regulation of stack temperature^[25]. Gruber *et al.* applied MPC to the PEMFC gas supply system and showed that satisfactory control performance can be achieved when an accurate system model is available^[26]. Liu *et al.* employed an enhanced MPC strategy for automotive fuel cell thermal management, significantly improving state equilibrium and fuel economy of onboard fuel cell systems^[27]. To further improve control performance, Yang *et al.* optimized an MPC controller using fuzzy logic, thereby enhancing system net power output and overall performance^[28]. Wang *et al.* combined MPC with PID control to improve the regulation performance of PEMFC gas supply systems^[29]. Chen *et al.* proposed a cascaded feedback linearization-based MPC scheme, which further strengthened the disturbance rejection capability of the system^[30]. More recently, Wang *et al.* introduced a deep learning-based hierarchical model predictive control (HMPC) framework, demonstrating improved robustness compared with conventional MPC approaches^[31]. Deng *et al.* developed a neural network-based MPC using long short-term memory (LSTM), with simulation results

indicating significant performance improvements^[32]. In addition, Chatrattanawet *et al.* proposed a robust MPC strategy for PEMFC control, with results confirming that the proposed method ensures system stability under uncertain conditions^[33].

However, model predictive control suffers from relatively long prediction horizons when applied to complex systems and requires considerable computational resources for online optimization, which may prolong the dynamic response time and consequently degrade the system's dynamic performance^[34]. Sliding mode control, on the other hand, may exhibit chattering during operation, and its performance is often constrained by model accuracy and operating conditions^[35]. To address these limitations and enhance the performance and stability of proton exchange membrane fuel cell systems, this paper integrates model predictive control with sliding mode control to develop an SMC-MPC fusion control strategy. Through numerical simulations, the proposed method is systematically compared with conventional single control strategies, thereby verifying the advantages of the SMC-MPC fusion controller in terms of dynamic performance and system stability.

The main contributions of this paper can be summarized as follows:

(1) A dynamic model of the PEMFC air supply subsystem is implemented based on established models in the literature, providing a simulation platform for control design and validation.

(2) A SMC-MPC hybrid control strategy is proposed, and its effectiveness is validated through numerical simulations.

The remainder of this paper is organized as follows. Section 2 presents the modeling of the PEMFC air supply subsystem and derives the relevant system parameters. Section 3 analyzes the SMC, MPC, and SMC-MPC control strategies. Section 4 evaluates the performance of the proposed controllers through simulation and discusses the corresponding results. Section 5 presents the conclusions of this paper. Finally, Section 6 discusses the limitations of this study and outlines directions for future research.

2. Modelling of PEMFC Air Supply Subsystem

2.1 Air Compressor Model

The air compressor model in the PEMFC air supply subsystem is a key component for accurately characterizing the dynamic behavior of cathode-side oxidant delivery. As the primary power unit of the air supply subsystem, the compressor directly affects the oxygen supply quality to the fuel cell stack and the overall system efficiency. In PEMFC applications, centrifugal compressors are commonly employed, typically operating at rotational speeds ranging from 30,000 to 100,000 rpm, with pressure ratios of 2.5 to 3.5.

From the motor torque balance equation, the dynamic equation for the air compressor speed is^[36]:

$$J_{cp} \frac{dw_{cp}}{dt} = t_{cm} - t_{cp} \quad (1)$$

Where J_{cp} is the rotational inertia of the air compressor, t_{cm} and t_{cp} are the motor torque and load torque.

The motor torque of the air compressor is:

$$t_{cm} = \frac{k_t h_{cm}}{R_{cm}} (V_{cm} - k_v w_{cp}) \quad (2)$$

Where k_t , k_v , R_{cm} are motor constants, h_{cm} is the mechanical efficiency of the motor, V_{cm} is the air compressor voltage, serving as the control variable characterizing the effectiveness of air compressor control, w_{cp} is the air compressor rotational speed.

Calculate the load torque using thermodynamic equations:

$$t_{cp} = \frac{c_p T_{atm}}{w_{cp} h_{cp}} \frac{\dot{e} \rho_{sm} \frac{g-1}{2g}}{\dot{e} \rho_{atm} \frac{g-1}{2g}} - 1 \dot{W}_{cp} \quad (3)$$

Where c_p is the specific heat capacity of air at constant pressure, $c_p = 1004 \text{ J/(kg}\cdot\text{K)}$, g is the specific heat coefficient, ρ_{atm} and ρ_{sm} are atmospheric pressure and supply line pressure, h_{cp} is the air compressor efficiency, T_{atm} is the atmospheric temperature, W_{cp} is the air compressor flow rate.

An air compressor model was constructed using the Simulink platform, as shown in Figure 1.

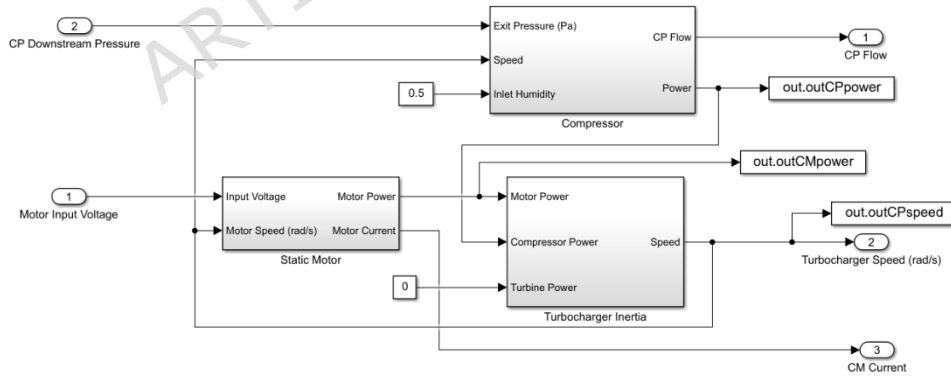


Figure 1. Air Compressor Model

2.2 Pipeline Transmission Model

The pipeline transport model of the PEMFC air supply subsystem is formulated to characterize the coupling mechanisms among pressure wave propagation, flow pulsation, and mass transfer. Based on the fundamental conservation laws of fluid mechanics, the model is constructed using a one-dimensional unsteady compressible flow governing equation system. The model incorporates key physical effects, including gas compressibility,

pipe wall friction, and heat exchange. the gas flow rate entering and exiting the cathode can be determined by the pressure difference between the compressor and the inlet of the supply pipe, from which it follows that:

$$\frac{dp_{sm}}{dt} = \frac{T_{cp}R}{V_{sm}M_a} (W_{CP} - K_{ca,in} (p_{sm} - p_{ca})) \quad (4)$$

$$T_{cp} = T_{atm} + \frac{T_{atm}}{h_{cp}} \left(\frac{p_{sm}}{p_{atm}} \right)^{\frac{g-1}{g}} - 1 \quad (5)$$

Where g is the thermal ratio coefficient for air, V_{sm} is the supply pipe volume, T_{cp} is the air compressor outlet temperature, $K_{ca,in}$ is the cathode inlet orifice constant.

A pipeline transmission model was constructed using the Simulink platform, as shown in Figure 2.

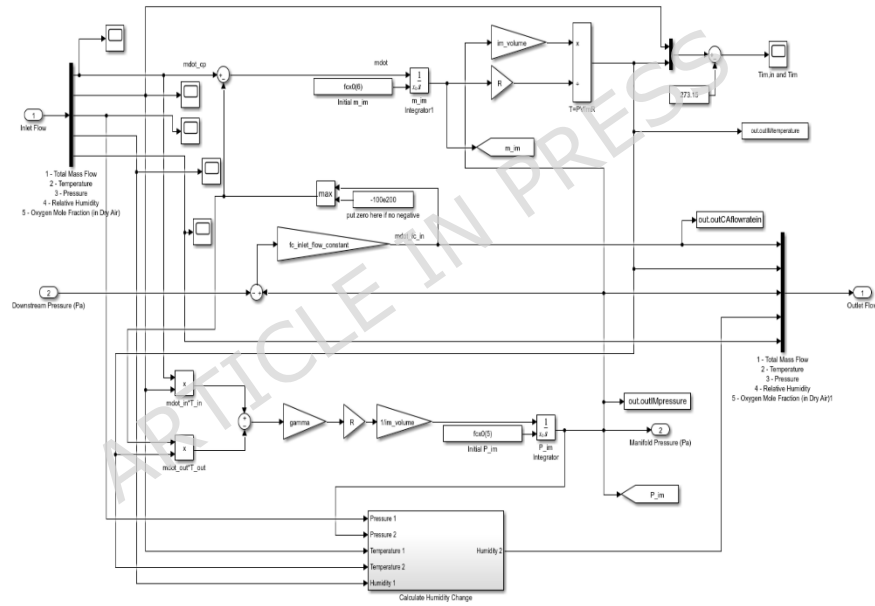


Figure 2. Pipeline Transmission Model

2.3 Cathode Reaction Model

The cathode reaction model of the PEMFC air supply subsystem is formulated using a multiscale approach to integrate macroscopic gas transport, mesoscopic pore diffusion, and microscopic electrochemical reactions. This model serves as the core mechanism for quantitatively describing the electrochemical reduction of oxygen at the cathode and its coupled interactions with gas transport processes.

Based on the law of conservation of mass, the dynamic equation for O_2 and N_2 flow rates can be expressed as:

$$\frac{dm_{O_2}}{dt} = W_{O_2,in} - W_{O_2,out} - W_{O_2,rc} \quad (6)$$

$$\frac{dm_{N_2}}{dt} = W_{N_2,in} - W_{N_2,out} \quad (7)$$

Where $W_{O_2,in}$ and $W_{N_2,in}$ are O_2 and N_2 flow rates into the cathode; oxygen also participates in reactions within the cathode, resulting in consumption, $W_{O_2,rc}$ is the flow rate of oxygen consumed by cathode reactions, $W_{O_2,out}$ and $W_{N_2,out}$ are the flow rates of O_2 and N_2 leaving the cathode without participating in reactions.

The O_2 and N_2 flow rates at the cathode inlet can be calculated based on the total flow rate at the cathode inlet and the proportion of O_2 and N_2 in the total gas mixture:

$$W_{O_2,in} = K_{O_2} \cdot W_{ca,in} \quad (8)$$

$$W_{N_2,in} = K_{N_2} \cdot W_{ca,in} \quad (9)$$

Where K_{O_2} and K_{N_2} are the proportion coefficients of O_2 and N_2 in the cathode input flow, while $W_{ca,in}$ denotes the total flow rate entering the cathode.

$$K_{O_2} = \frac{1}{1+W_{atm}} y_{O_2} \quad (10)$$

$$K_{N_2} = \frac{1}{1+W_{atm}} y_{N_2} \quad (11)$$

$$y_{O_2} = \frac{c_{O_2} M_{O_2}}{M_a} \quad (12)$$

$$y_{N_2} = \frac{(1 - c_{O_2}) M_{N_2}}{M_a} \quad (13)$$

Where y_{O_2} and y_{N_2} are the mass fractions of O_2 and N_2 , respectively; W_{atm} is the humidity ratio at standard atmospheric pressure, c_{O_2} is the volume fraction of oxygen in air, M_{O_2} and M_{N_2} are the molar masses of O_2 and N_2 , M_a is the molar mass of air.

Since the content of water vapor and other gases in the cathode input gas is negligible, it is assumed that the input gas consists solely of O_2 and N_2 . Set $c_{O_2} = 21\%$.

The cathode gas inlet flow rate can be determined by the pressure difference between the supply pipe and the cathode gas pressure:

$$W_{ca,in} = \frac{1}{1+W_{atm}} W_{in} \quad (14)$$

$$W_{in} = K_{ca,in} (p_{sm} - p_{ca}) \quad (15)$$

$$W_{atm} = \frac{M_v}{M_a} \frac{f_{atm} P_{sat}}{P_{atm} f_{atm} P_{sat}} \quad (16)$$

Where f_{atm} is the ambient relative humidity, taken as 0.5; P_{atm} and P_{sat} are atmospheric pressure and saturated vapor pressure. $K_{ca,in}$ is the orifice plate constant at the cathode outlet.

Derive the cathode outlet flow rate based on the relationship between cathode pressure and atmospheric pressure difference. The corresponding outlet flow rates of O_2 and N_2 are determined by their respective pressure ratios within the system:

$$W_{ca,out} = K_{ca,out} \sqrt{P_{ca} + P_{atm}} \quad (17)$$

$$W_{N_2,out} = \frac{P_{N_2}}{P_{ca}} W_{ca,out} \quad (18)$$

$$W_{O_2,out} = \frac{P_{O_2}}{P_{ca}} W_{ca,out} \quad (19)$$

The mass flow rate of O_2 consumption is:

$$W_{O_2,rc} = M_{O_2} \frac{nI}{4F} \quad (20)$$

Where n is the number of individual cells comprising the stack, I is the stack current.

The pressure of the cathode gases O_2 and N_2 is:

$$P_{O_2} = \frac{RT_{st}}{M_{O_2} V_{ca}} m_{O_2} \quad (21)$$

$$P_{N_2} = \frac{RT_{st}}{M_{N_2} V_{ca}} m_{N_2} \quad (22)$$

$$P_{ca} = P_{O_2} + P_{N_2} + P_{sat} \quad (23)$$

Where V_{ca} is cathode volume, T_{st} is stack temperature, P_{sat} is water vapor pressure.

A cathode air supply system was constructed using the Simulink platform, as shown in Figure 3.

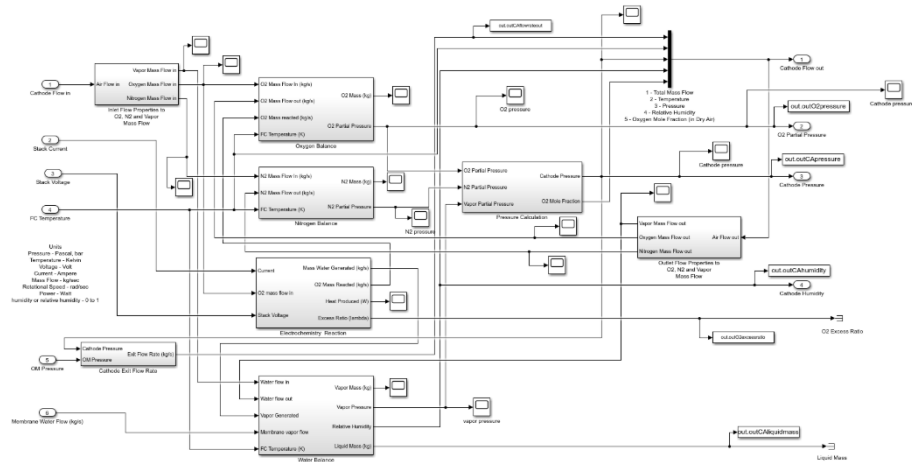


Figure 3. Cathode Air Supply System

2.4 Reflux Pipeline Model

According to the law of conservation of mass, the differential equation for the reflux tube pressure can be derived as follows^[36]:

$$\frac{dp_{rm}}{dt} = \frac{R_a T_{rm}}{V_{rm}} (W_{ca,out} - W_{rm,out}) \quad (24)$$

Where W_{rmout} , V_{rm} , T_{rm} are the flow rate, volume, and temperature of the reflux tube, respectively.

The nonlinear nozzle equation yields:

$$W_{mout} = \begin{cases} \frac{C_D A_T P_{rm}}{\sqrt{RT_{rm}}} (PR)^{\frac{1}{g}} \frac{2g}{g-1} \left(\frac{P_{atm}}{P_{rm}} \right)^{\frac{g-1}{g}}, & PR = \frac{P_{atm}}{P_{rm}} > PR_{crit} \\ \frac{C_D A_T P_{rm}}{\sqrt{RT_{rm}}} (PR)^{\frac{1}{g}} \frac{2}{g-1} \left(\frac{P_{atm}}{P_{rm}} \right)^{\frac{g+1}{2(g-1)}}, & PR = \frac{P_{atm}}{P_{rm}} < PR_{crit} \end{cases} \quad (25)$$

Critical pressure drop PR_{cnt} is defined as:

$$PR_{cnt} = \frac{2}{g+2} \frac{g}{g-1} \quad (26)$$

The meanings and values of the parameters used in the above modeling process are shown in Table 1.

Table 1. Parameters in the Model Development Process for the PEMFC Air Supply System

Parameters	Physical Meaning	Values and Units
T	Operating temperature	353K
M_a	Air mole mass	29×10^{-3} kg/mol
M_{O_2}	Molar mass of oxygen	32×10^{-3} kg/mol

M_{H_2}	Molar mass of hydrogen	$2 \cdot 10^{-3} \text{ kg/mol}$
M_{N_2}	Molar mass of nitrogen	$28 \cdot 10^{-3} \text{ kg/mol}$
$K_{ca,out}$	Cathode outlet flow coefficient	$1.28 \cdot 10^{-6} \text{ mol/(sPa)}$
P_b	Export pressure	$1.01 \cdot 10^5 \text{ Pa}$
J_{cp}	Compressor rotational inertia	$671.9 \cdot 10^{-5} \text{ kg m}^2$
k_t	Motor constants	0.31 V/A
k_v	Motor constants	$0.0153 \text{ V/(rad sec)}$
h_{cp}	Air compressor efficiency	80%
h_{cm}	Mechanical efficiency of electric motors	98%
g	Thermal proportionality coefficient	1.4

2.5 Derivation of the State Equation

Based on the formula derivations for the fuel cell gas supply system described above, a fourth-order state equation for the PEMFC air supply system is established. The four state variables x_1 , x_2 , x_3 , and x_4 are the cathode oxygen pressure P_{O_2} , cathode nitrogen pressure P_{N_2} , compressor speed W_{cp} , and supply line pressure P_{sm} , respectively. The stack load current I_{st} is represented by x as the disturbance term for the system input, u is the system input, which in this case denotes the compressor voltage V_{cm} .

$$\begin{aligned} \dot{x}_1 = & \frac{RT_{st}K_{ca,in}K_{O_2}}{M_{O_2}V_{ca}(1+w_{atm})}(x_4 - x_1 - x_2 - p_{sat}) \\ & - \frac{RT_{st}K_{ca,out}}{V_{ca}M_{O_2}}\left(\frac{x_1}{x_1+x_2+p_{sat}}\right)\sqrt{x_1+x_2+p_{sat}-P_{atm}} - \frac{RT_{st}K_{ca,out}}{4V_{ca}F}z \end{aligned} \quad (27)$$

$$\begin{aligned} \dot{x}_2 = & \frac{RT_{st}K_{ca,in}K_{O_2}}{M_{O_2}V_{ca}(1+w_{atm})}(x_4 - x_1 - x_2 - p_{sat}) \\ & - \frac{RT_{st}K_{ca,out}}{V_{ca}M_{O_2}}\left(\frac{x_2}{x_1+x_2+p_{sat}}\right)\sqrt{x_1+x_2+p_{sat}-P_{atm}} \end{aligned} \quad (28)$$

$$\dot{x}_3 = \frac{-h_{cm}K_tK_v}{J_{cp}R_{cm}}x_3 - \frac{C_pT_{atm}}{h_{cp}}\frac{\frac{\partial x_4}{\partial P_{atm}}\frac{\partial g}{\partial P_{atm}} - 1}{g}W_{cp} + \frac{h_{cm}K_t}{J_{cp}R_{cm}}u \quad (29)$$

$$\dot{x}_4 = \frac{R}{V_{sm}}\left[T_{atm} + \frac{T_{atm}}{h_{cp}}\frac{\frac{\partial x_4}{\partial P_{atm}}\frac{\partial g}{\partial P_{atm}} - 1}{g}W_{cp} - K_{ca,in}(x_4 - x_1 - x_2 - p_{sat})\right] \quad (30)$$

There exists a functional relationship between the oxygen excess ratio and the net power output. In most existing PEMFC air supply system control strategies, the oxygen excess ratio is typically maintained at a fixed value of 2. However, in practical operation, the optimal oxygen excess ratio that maximizes net power varies with changes in load current. To achieve improved control performance, the oxygen excess ratios corresponding to maximum net power under different load currents are summarized in Table 2.

Table 2. Peroxide Ratio for Maximum Netpower at Different Load Currents

$I_{st}(A)$	100	120	140	160	180	200	220	240
$\lambda_{O_2}^{opt}$	2.429	2.395	2.364	2.318	2.297	2.263	2.219	2.197

By performing a linear fit on the data in Table 2, the relationship between the optimal peroxide ratio $\lambda_{O_2}^{opt}$ and the load current is:

$$\lambda_{O_2}^{opt} = -0.0017 I_{st} + 2.5973 \quad (31)$$

Based on the above analysis of the control system and control objectives, the nonlinear system model of the PEMFC is:

$$\begin{cases} \dot{x} = f(x, u, z) \\ x = [P_{O_2}, P_{N_2}, W_{cp}, P_{sm}]^T \\ y = [\lambda_{O_2}, P_{net}]^T \\ u = V_{cm} \\ z = I_{st} \end{cases} \quad (32)$$

3. Control Strategy

3.1 Sliding Mode Control Strategy

3.1.1 Calculation of The Sliding Surface $S(x)$

The switching function $S(x)$ for SMC is:

$$S(x) = W_{cp} - W_{cp,ref} \quad (33)$$

Where W_{cp} is the actual flow rate of the air compressor, and $W_{cp,ref}$ is the desired flow rate of the air compressor.

The control objective of the air supply system is to maintain the excess air ratio at its optimal value at all times; therefore, it is necessary to derive the relationship between the desired flow rate $W_{cp,ref}$ and $\lambda_{O_2,ref}$:

$$W_{dryair,ref} = \frac{1}{C_{O_2}} W_{O_2,inref} \quad (34)$$

$$/_{O_2,ref} = \frac{W_{O_2,inref}}{W_{O_2,rctref}} \quad (35)$$

$$W_{O_2,inref} = /_{O_2,ref} W_{O_2,rctref} \quad (36)$$

Where $W_{O_2,inref}$ is the reference oxygen consumption, $W_{O_2,rctref}$ is the actual oxygen consumption, and $/_{O_2,ref}$ is the reference excess air ratio.

$$W_{O_2,inref} = M_{O_2} \frac{nI_{st}}{4F} \quad (37)$$

Where M_{O_2} is the oxygen input, I_{st} is the load current, n and F are constants.

Based on the above formula, the dry air flow rate at the reference excess oxygen ratio can be derived as follows:

$$W_{dryair,ref} = \frac{1}{C_{O_2}} /_{O_2,ref} M_{O_2} \frac{nI_{st}}{4F} \quad (38)$$

Where $W_{dryair,ref}$ is the dry air flow rate and $/_{O_2}$ is the excess air ratio.

Consider the effect of relative humidity on the air compressor flow rate:

$$W_{cp,ref} = (1+W_{atm}) \frac{1}{C_{O_2}} /_{O_2,ref} M_{O_2} \frac{nI_{st}}{4F} \quad (39)$$

Where W_{atm} is the humidity correction factor at atmospheric pressure, C_{O_2} is the molar fraction of oxygen in the air, $/_{O_2,ref}$ is the reference value for the excess oxygen ratio.

The relationship between $W_{cp,ref}$ and the load current is:

$$W_{cp,ref} = (2.5973 - 0.0017I_{st})(1+W_{atm}) \frac{1}{C_{O_2}} M_{O_2} \frac{nI_{st}}{4F} \quad (40)$$

3.1.2 Selection of Control Function $u(x)$

$$\begin{aligned} S(x, t) &= W_{cp} - W_{cp,ref} \\ &= A \left[1 - e^{B \left(\left(\frac{x_4}{P_{atm}} \right) - 1 \right) x_3^2 - C} \right] x_3 - (1+W_{atm}) \frac{1}{C_{O_2}} W_{O_2,inref} /_{O_2,ref} M_{O_2} \frac{nI_{st}}{4F} \end{aligned} \quad (41)$$

$$\begin{aligned} \dot{S}(x, t) &= A \dot{x}_3 \left[1 - e^{B \left(\left(\frac{x_4}{P_{atm}} \right) - 1 \right) x_3^2 - C} \right] \\ &\quad - 3ABx_3^2 A \frac{\partial}{\partial x_3} e^{B \left(\left(\frac{x_4}{P_{atm}} \right) - 1 \right) x_3^2 - C} \frac{\partial}{\partial x_4} \left(\frac{x_4}{P_{atm}} \right) - 1 \frac{\partial}{\partial x_3} \dot{x}_3 \\ &\quad + \frac{A x_4 x_3}{P_{atm}} \frac{\partial}{\partial x_3} e^{B \left(\left(\frac{x_4}{P_{atm}} \right) - 1 \right) x_3^2 - C} \frac{\partial}{\partial x_4} \dot{x}_4 + D \frac{dI_{st}}{dt} \end{aligned} \quad (42)$$

$$\dot{x}_3^g = \frac{-h_{cm}K_tK_v}{J_{cp}R_{cm}}x_3 - \frac{C_pT_{atm}}{h_{cp}} \frac{\frac{\partial x_4}{\partial g} \frac{g-1}{g}}{J_{cp}x_3} W_{cp} + \frac{h_{cm}K_t}{J_{cp}R_{cm}}u \quad (43)$$

By taking the first derivative of $S(x, \dot{x})$, the control input $u(x)$ can be obtained. Analysis of the above derivation indicates that when the system is governed by a first-order sliding surface, oscillations may arise during control execution. In the context of an air compressor system, such voltage oscillations can adversely affect motor performance and may even lead to potential damage. Therefore, it is essential to explicitly consider and mitigate oscillatory behavior during controller design to ensure system stability and the safe operation of the motor.

Advanced sliding mode control strategies typically require detailed knowledge of the derivative characteristics of the switching function, which significantly increases both analytical and computational complexity. In contrast to several commonly used second-order sliding mode methods, the hyper-helix algorithm does not require detailed derivative information of S , but relies only on the value of S itself. For first-order systems, this method enables direct implementation of the control strategy, with a relatively intuitive and straightforward structure that involves only three key parameters. Compared with conventional sliding mode control methods, it is more effective in suppressing oscillatory behavior.

Therefore, the hyper-helix algorithm is adopted in this study to regulate the PEMFC air supply system. Accordingly, the control law is defined as follows:

$$\begin{cases} u = -1/|S|^r \operatorname{sign}(S)u_1 \\ \dot{u}_1^g = -a \operatorname{sign}(S) \end{cases} \quad (44)$$

where sign is a sign function, and the value of a and r is:

$$\begin{cases} a > \frac{H}{K_m} \\ |S|^2 > 2 \frac{aK_m + H}{K_m} \\ 0 \leq r \leq 0.5 \end{cases} \quad (45)$$

3.2 Model Predictive Control Strategy

3.2.1 Dynamic Matrix Control Model

Dynamic Matrix Control (DMC) is a representative model predictive control technique whose fundamental principle is based on the system's step response model to construct a closed-loop control framework. Compared with conventional predictive control methods,

DMC features a relatively simple implementation procedure, strong robustness, fast dynamic response, and low dependence on model accuracy.

The core of the DMC predictive control model lies in constructing the system's step response curve to determine an appropriate sampling period T . This model is primarily intended to satisfy the control requirements of the system under steady-state operating conditions, and it utilizes step response data over N consecutive sampling instants to characterize the system's dynamic behavior, where N denotes the time-domain length parameter of the model.

From the step response curve, the sampled values $a_1, a_2, a_3, \dots, a_N$ are obtained at each sampling instant from 1 to N . Based on this sequence of step response coefficients, future output values can be predicted. The corresponding change in $y(k)$ can be expressed as:

$$\Delta y(k) = \begin{cases} a_i \Delta u(k-i), & (1 \leq i < M) \\ a_N \Delta u(k-i), & (i \geq M) \end{cases} \quad (46)$$

By the superposition theorem, the output value $y(k)$ is given by:

$$y(k) = \sum_{i=1}^{N-1} a_i \Delta u(k-i) + a_N \Delta u(k-M) \quad (47)$$

From the above equation, we can see that when $n < N$, the estimated value at time step a is:

$$\hat{y}(k+j) = \sum_{i=1}^{N-1} a_i \Delta u(k+j-i) + a_N \Delta u(k+j-M) \quad (j=1, 2, \dots, a) \quad (48)$$

The calculation of the estimated value in the above equation takes into account the influence of inputs; when the inputs are known, the equation is solved separately.

$$\hat{y}(k+j) = \sum_{i=1}^{N-1} a_i \Delta u(k+j-i) + \sum_{i=j+1}^{N-1} a_i \Delta u(k+j-1) + a_N \Delta u(k+j-M) \quad (j=1, 2, \dots, a) \quad (49)$$

The last two terms in the above equation represent a -step predictions of the output, denote these two terms as:

$$y_0(k+j) = \sum_{i=j+1}^{N-1} a_i \Delta u(k+j-1) + a_N \Delta u(k+j-M) \quad (j=1, 2, \dots, a) \quad (50)$$

$$\begin{aligned} \hat{y}_0(k+1) &= a_1 \Delta u(k) \\ \hat{y}_0(k+2) &= a_1 \Delta u(k) + a_2 \Delta u(k+1) \\ &\vdots \\ \hat{y}_0(k+a) &= a_1 \Delta u(k) + a_2 \Delta u(k+1) + \dots + a_{a-1} \Delta u(k+a-1) \end{aligned} \quad (51)$$

To reduce the computational load while improving the controllability and dynamic stability of the system, we reduce the dimension to m ($m < a$), in which case (51) can be simplified to:

$$\begin{bmatrix} y_0(k+1) \\ y_0(k+2) \\ \vdots \\ y_0(k+a) \end{bmatrix} = \begin{bmatrix} a_1 & 0 & \dots & 0 \\ a_2 & a_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_a & a_{a-1} & \dots & a_{a-m+1} \end{bmatrix} \begin{bmatrix} Du(k) \\ Du(k+1) \\ \vdots \\ Du(k+m-1) \end{bmatrix} + \begin{bmatrix} y_0(k+1) \\ y_0(k+2) \\ \vdots \\ y_0(k+a) \end{bmatrix} \quad (52)$$

Equation (52) can be rewritten as:

$$Y = AU + Y_0 \quad (53)$$

All elements of matrix A can be obtained from the system's step response curve, and it is an $a \times m$ -dimensional matrix. As a dynamic matrix, A establishes the relationship between input increments and future outputs. According to the above formulation, given the system output at time K , the future output over the prediction horizon can be determined based on the input increments and the dynamic matrix.

3.2.2 Dynamic Matrix Scrolling Optimization

Since large sudden changes in the control input are undesirable during the control process, an optimization criterion is used to constrain the control increment, the formula for the optimization criterion is:

$$J = \sum_{j=1}^g \alpha(j) [y(k+1) - u(k+j)]^2 + \sum_{j=1}^m \lambda(j) [Du(k+j-1)]^2 \quad (54)$$

Where $\alpha(j)$ is the tracking error suppression coefficient, $\lambda(j)$ is the incremental change suppression coefficient.

The sequence of $u(k+j)$ in the formula represents the expected output:

$$u(k+j) = a^j y(k) + (1 - a^j) y^*(j=1, 2, \dots, n) \quad (55)$$

Equation (54) can be expressed using vector notation:

$$J = \alpha(Y - W)(Y - W)^T + \lambda DU^T DU \quad (56)$$

Replace Y with the optimal forecast value $\hat{Y}$ and let $\frac{\partial J}{\partial U} = 0$ can be obtained:

$$DU = (A^T A + \lambda I)^{-1} A^T (W - Y_0) \quad (57)$$

$$Du(k) = c^T (A^T A + \lambda I)^{-1} A^T (W - Y_0) \quad (58)$$

As indicated by the above derivation, once the control input $u(k)$ is implemented, predictive control requires real-time acquisition of output data at subsequent time steps and continuously performs prediction, correction, and optimization. In contrast to traditional offline optimization methods, the optimization process in dynamic matrix control is inherently online and recursive. Its objective function is optimized in a receding horizon manner, exhibiting local optimality at each time step, with the optimal solution obtained through iterative computation. This dynamic updating mechanism reflects the principle of rolling optimization in DMC, whereby system performance is progressively improved through

continuous adjustment, rather than relying on a fixed global optimization over the entire time horizon.

3.2.3 Dynamic Matrix Feedback Correction

In each control cycle, only the first control increment $u(k)$ is used to predict the output at some future time using:

$$\dot{Y}_p = aDu(k) + \dot{Y}_{p0} \quad (59)$$

$\dot{Y}_p = [\dot{y}_p(k+1), \dot{y}_p(k+2), \dots, \dot{y}_p(k+p)]^T$ is the system output at the next p time steps, as predicted

at time $t=kT$, assuming the effect of $Du(k)$, $\dot{Y}_{p0} = [\dot{y}_0(k+1), \dot{y}_0(k+2), \dots, \dot{y}_0(k+p)]^T$ is the system output predicted at time $t=kT$ for the next p time steps, assuming no $Du(k)$ is applied, $a = [a_1, a_2, \dots, a_p]$ is the values obtained at each sampling time under the system's unit step response.

By calculation, the predicted output value at time step $k+1$ is:

$$\dot{y}(k+1) = \dot{y}_0(k+1) + a_1 Du(k) \quad (60)$$

Due to uncertainties in the control process, after a control action is applied at time k , the actual output $y(k+1)$ at time $k+1$ may deviate from the predicted value. The discrepancy between the predicted output and the actual output can be expressed as:

$$e(k+1) = y(k+1) - \dot{y}(k+1) \quad (61)$$

Using the obtained error values to correct the output values for future time steps yields:

$$\dot{Y}_p = \dot{Y}_{p0} + h e(k+1) \quad (62)$$

The calibrated $\dot{Y}_p$ is used as the initial value for the next prediction. Given the initial value at time $t=(k+1)T$, the output values at time $t=(k+2)T, \dots, (k+p+1)T$ should be predicted.

$$\dot{y}_0(k+i) = \dot{y}(k+i+1) \quad (i=1, 2, \dots, p-1) \quad (63)$$

From the above derivation, we can see that the predicted initial value for the next time step is:

$$\dot{y}_0(k+i) = \dot{y}(k+i+1) + h_{i+1} e(k+1) \quad (i=1, 2, \dots, p-1) \quad (64)$$

$$\dot{y}_0(k+p) = \dot{y}(k+p) + h_p e(k+1) \quad (65)$$

This optimization scheme establishes a closed-loop feedback control framework. Its key feature lies in the incorporation of a feedback correction mechanism, which enhances system

stability and significantly improves the system's ability to adapt to external disturbances, as well as its robustness against model uncertainties.

3.2.4 Linearization of the PEMFC Air Supply System Model

Due to the high complexity of the nonlinear model of the fuel cell gas supply system, it cannot be directly applied within conventional model predictive control frameworks. Therefore, model linearization is performed to simplify the system representation. In this study, the air compressor voltage V_{am} is selected as the system input, and the oxygen excess ratio λ_{O_2} is chosen as the system output for linearization. When the load current is 100A and the oxygen excess ratio is 2.43, the corresponding system state is denoted as $x_e = [0.2174, 1.104, 49.95, 1.427]$, which is taken as the nominal operating point of the system. By performing a Taylor series expansion of equation (32) around the operating point x_e , a fourth-order state-space representation of the system is obtained. The parameters of the resulting fourth-order state-space model are given as follows:

$$A = \begin{bmatrix} -9.397 & -7.417 & 0 & 7.623 \\ 28.29 & -30.55 & 0 & 29.17 \\ 0 & 0 & -5.78 & 0.1366 \\ 23.92 & 23.92 & 65.79 & -24.42 \end{bmatrix} \quad (66)$$

$$B = [0 \ 0 \ 367.5 \ 0] \quad (67)$$

$$C = [-0.0002631 \ -0.0002631 \ 0 \ 0.0002631] \quad (68)$$

$$D = [0] \quad (69)$$

Thus, the transfer function of the system is:

$$G(s) = \frac{6.361s^2 + 19.68s + 13.66}{s^4 + 70.13s^3 + 552.3s^2 + 1439s + 1214} \quad (70)$$

Using MATLAB, the eigenvalues of matrix A are:

$$[-61.5252 \ -4.2962 \ -2.0186 \ -2.3069] \quad (71)$$

Since all eigenvalues of matrix A have negative real parts, it follows from Lyapunov stability theory that the linearized system is locally stable at the selected operating point.

3.3 SMC-MPC Hybrid Control Strategy

3.3.1 Principles of SMC-MPC Hybrid Control Strategy

The fundamental concept of the SMC-MPC hybrid controller is summarized as follows:

1. The MPC component is responsible for global optimization by predicting the future behavior of the system and generating an optimal control sequence, thereby facilitating operation within prescribed constraints.

2. The SMC component provides fast local compensation to mitigate the effects of external disturbances and model uncertainties, thereby enhancing the robustness of the overall control system.

The outputs are subsequently integrated using a fusion algorithm. Common fusion strategies include cascade structure, weighted fusion, and conditional switching. In the cascade structure, the MPC generates the reference trajectory, while the SMC is responsible for tracking this trajectory. In weighted fusion, the control outputs of the two controllers are combined proportionally. In conditional switching, MPC is employed during steady-state operation, whereas the controller switches to SMC when disturbances occur. In this study, a cascaded control structure is adopted, the overall control block diagram is presented in Figure 4.

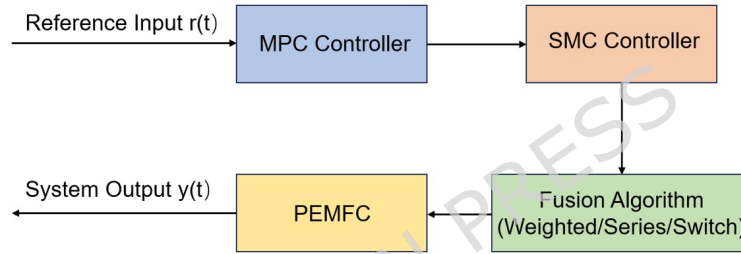


Figure 4. Principle of Fusion Control Strategy

3.3.2 MPC Module Design

The fundamental principle of Model Predictive Control is to employ a predictive model of the system to forecast its future outputs over a specified prediction horizon, thereby obtaining an optimal control input sequence through a rolling optimization process. In this study, the MPC is implemented within the DMC framework. Based on the DMC formulation, a fourth-order state-space representation of the system is derived. The cathode oxygen partial pressure, nitrogen partial pressure, manifold pressure, and compressor speed are selected as the state variables. After discretization, the control equations can be expressed as follows:

$$\begin{cases} \dot{x}(k+1) = Ax(k) + Bu(k) \\ y(k) = Cx(k) \end{cases} \quad (72)$$

Where $x(k)$ is the state variable vector representing the internal state of the system at time k . Since the cathode oxygen partial pressure, nitrogen partial pressure, manifold pressure, and compressor speed are selected as state variables, $x(k)$ is a 4×1 vector. $u(k)$ is the input vector representing the control quantities applied to the system at time k , such

as compressor control voltage and power, which influence air flow rate, pressure, and oxygen concentration by driving the compressor, $y(k)$ is the output vector, representing the observable quantities of the system at time k , such as stack voltage, net power, or cathode oxygen over-ratio; The matrix A is a 4×4 state transition matrix, reflecting the dynamic coupling relationships among oxygen partial pressure, nitrogen partial pressure, manifold pressure, and compressor speed. It determines how the system state evolves over time in the absence of input; The matrix B is a 4×1 input matrix describing the influence of input $u(k)$ on state variables; The matrix C is a 4×1 output matrix describing how state variables map to outputs.

MPC controller prediction and control time domain:

$$N_p = 10 \quad (73)$$

$$N_c = 3 \quad (74)$$

$$Dt = 0.1s \quad (75)$$

Where N_p is the prediction time domain, N_c is the control time domain, Dt is the step size. A prediction step size of $N_p = 10$ is used to capture the system's primary dynamic processes, while a control step size of $N_c = 3$ is employed to reduce the dimensionality of the online optimization and improve computational efficiency. The sampling time $Dt = 0.1s$ is determined based on the system's dynamic response characteristics and real-time control requirements. These parameters were determined through comparative simulation and debugging across multiple sets of experiments, achieving a good balance between performance and computational complexity.

Constraints for the MPC Controller: This paper treats air flow rate and compressor speed as the primary controlled variables, with their constraints expressed as follows:

$$0 \leq \dot{m}_{air}(t) \leq \frac{0.02kg}{s} \quad (76)$$

$$2.09 \times 10^3 \frac{rad}{s} \leq N_{comp} \leq 1.26 \times 10^4 \frac{rad}{s} \quad (77)$$

The upper and lower limits for airflow are set to $0 \sim 20g/s$, with the compressor's minimum rotational speed at $2.09 \times 10^3 rad/s$ and maximum rotational speed at $1.26 \times 10^4 rad/s$. This constraint configuration ensures sufficient oxygen supply capacity for the fuel cell under dynamic operating conditions while preventing compressor overload during high-speed operation. It thus balances system safety, control feasibility, and operational efficiency.

The design objective of the MPC is formulated as a quadratic objective function that quantifies both the tracking performance of the system output and the variation of the control

inputs. The optimization procedure seeks to determine the optimal control input sequence by minimizing this objective function, which can be decomposed into the following components:

1. Minimization of output tracking error:

The objective function enforces accurate tracking of the reference trajectory by minimizing the sum of squared deviations between the system output and the reference signal. The first term of the objective function represents the output tracking error:

$$J_1 = \sum_{i=1}^{N_p} (y(k+i) - y_{ref}(k+i))^2 \quad (78)$$

Where $y(k+i)$ is the system output at step i (e.g., partial pressure of oxygen, oxygen-to-peroxide ratio); $y_{ref}(k+i)$ is the reference trajectory (i.e., target value).

2. Minimization of control increments:

This objective is intended to suppress abrupt variations in the control inputs, thereby ensuring smooth control actions and avoiding unnecessary sharp fluctuations. Minimizing control increments contributes to reduced energy consumption associated with control actions and improves the overall operational efficiency of the system. The second term of the objective function represents the control increment:

$$J_2 = \lambda \sum_{i=0}^{N_c-1} (\Delta u(k+i))^2 \quad (79)$$

Where $\Delta u(k+i)$ is the increment for the step i control input, λ is the weighting factor for the control increment. This parameter is used to balance tracking performance and the smoothness of the control input; its value was determined through multiple rounds of simulation debugging. During the parameter tuning process, it was observed that when λ is small ($\lambda < 0.2$), the system response is fast but the control input exhibits significant fluctuations; conversely, when λ is large ($\lambda > 1$), the system response becomes sluggish. After comprehensively considering dynamic performance and actuator behavior, $\lambda = 0.5$ was ultimately selected.

In summary, the objective function of the MPC module is:

$$J = J_1 + J_2 = \sum_{i=1}^{N_p} (y(k+i) - y_{ref}(k+i))^2 + \lambda \sum_{i=0}^{N_c-1} (\Delta u(k+i))^2 \quad (80)$$

The primary control objective of the MPC module is to enhance system performance by minimizing both the tracking error and the control increment, while explicitly accounting for system constraints during the optimization process. Through rolling optimization and real-time computation, the MPC adaptively adjusts the control inputs, thereby ensuring stable and efficient operation of the PEMFC air supply subsystem under dynamic operating conditions.

3.3.3 SMC Module Design

The key aspect of sliding mode controller design lies in the formulation of the sliding surface, which is constructed using state variables associated with the tracking error and its rate of change. Based on the state variable matrix defined in this study, the sliding surface is expressed as follows:

$$s(t) = c \times e(t) + \dot{e}(t) \quad (81)$$

Where $e(t)$ is the tracking error, defined as the difference between the reference value and the actual value:

$$e(t) = I_{ref}(t) - I(t) \quad (82)$$

Where $I_{ref}(t)$ is the reference oxygen ratio, $I(t)$ is the actual output oxygen ratio, c is the proportional gain, used to amplify the effect of error on control, is typically set to 5, $\dot{e}(t)$ is the error derivative, representing the rate of change of error.

The control law of the sliding mode controller is designed using the boundary layer method to mitigate chattering phenomena in the system. The control input is formulated as follows:

$$u(t) = u_{eq}(t) - k \times \text{sat}_{\frac{\dot{e}(t)}{f}} \frac{\dot{e}(t)}{f} \quad (83)$$

Where $u_{eq}(t)$ is the equivalent control term, corresponding to the optimal control input computed by the MPC module, k is the switching gain, which characterizes the response intensity of the sliding mode controller to disturbances. A larger value of k implies stronger robustness, and in this study it is set to $k=10$. The function $\text{sat}_{\frac{\dot{e}(t)}{f}} \frac{\dot{e}(t)}{f}$ is the saturation function, which is introduced to constrain the control input and prevent excessive control actions. The parameter f is the boundary layer thickness, which defines the transition region near the sliding surface to avoid chattering caused by high-frequency switching. In this paper, the boundary layer thickness is chosen as $f=0.02$.

3.3.4 SMC-MPC Converged Controller

In the SMC-MPC hybrid control strategy, the SMC module is employed for local compensation by rapidly responding to disturbances and system uncertainties, thereby enhancing overall robustness. Meanwhile, the MPC module is responsible for global optimization, predicting future system behavior and computing the optimal reference trajectory and control inputs over the prediction horizon while satisfying system constraints. Specifically:

1. The optimal control inputs generated by the MPC module are obtained through a global optimization process that accounts for the predicted future behavior of the system and produces a sequence of control actions over the prediction horizon.

2. Based on the instantaneous tracking error and its derivative, the SMC module rapidly compensates for the reference trajectory provided by the MPC through sliding mode control, thereby suppressing disturbances and system uncertainties and enhancing overall robustness. The specific control signals are combined using a cascade structure:

Based on the objective function (80) and considering the associated constraints, the following expression is obtained:

$$u_{MPC}(t) = \underset{u}{\operatorname{argmin}} \left(\sum_{i=1}^{N_p} (y(k+i) - y_{ref}(k+i))^2 + \lambda \sum_{i=0}^{N_c-1} (Du(k+i))^2 \right) \quad (84)$$

Where argmin is the variable value that minimizes the objective function.

By applying the sliding mode control law given in (83) for rapid compensation, the following result is obtained:

$$u_{SMC}(t) = -k_{sat} \frac{s(t)}{f} \quad (85)$$

By combining (84) and (85), the following expression is obtained:

$$u(t) = u_{MPC}(t) + u_{SMC}(t) \quad (86)$$

Where $u_{MPC}(t)$ is the control signal computed by the MPC module; $u_{SMC}(t)$ is the compensation signal provided by the SMC module.

Specifically, the MPC module provides a predictive and constraint-handling control action based on the system model, ensuring optimal performance under nominal conditions. The SMC module is then introduced to compensate for modeling errors and disturbances, thereby improving tracking accuracy and robustness. The coupling between the two control modules enables the system to not only accurately track the reference trajectory but also rapidly adapt to external disturbances, thereby ensuring stable operation under complex operating conditions.

In this study, the oxygen excess ratio is selected as the primary controlled variable, as it directly reflects the balance between oxygen supply and consumption in the PEMFC system. Meanwhile, variables such as cathode pressure and compressor speed are indirectly regulated to ensure overall system stability and performance. The air compressor voltage is chosen as the primary manipulated variable, since it governs the compressor dynamics and determines the air flow rate delivered to the cathode. Through coordinated adjustment of this control input, the proposed SMC-MPC strategy achieves accurate tracking of the desired oxygen excess ratio while maintaining stable system operation under dynamic load variations.

4. Results and discussion

4.1 Analysis of Sliding Mode Control Simulation Results for PEMFC Gas Supply Subsystem

In this study, a sliding mode control model for the PEMFC system is established using Simulink simulation software. Set $a=2, l=3, r=0.5$ in the sliding mode controller. Under the sliding mode control strategy, the temporal responses of the load current, switching function S , output voltage, oxygen excess ratio, and net power are systematically analyzed. The load current input for the SMC is shown in Figure 5.

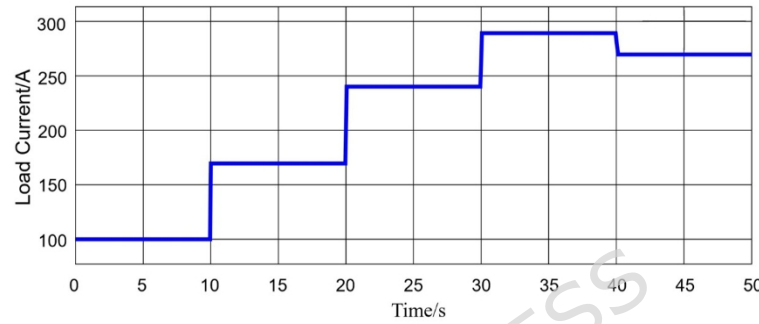


Figure 5. Load Current Input for SMC

Figure 6 illustrates the response of the switching function to variations in the load current. Following a change in the load current, the switching function exhibits rapid oscillations and converges to the sliding surface within less than 1s. Subsequently, it evolves smoothly along the sliding surface until the next load current variation occurs, at which point it readjusts to regulate the system outside the sliding surface.

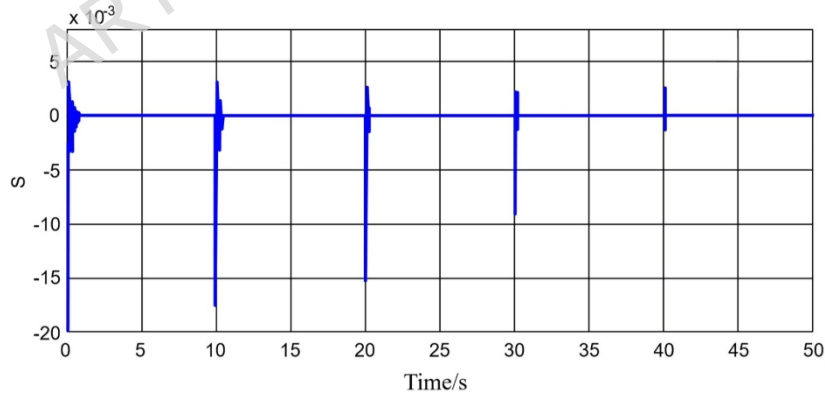


Figure 6. Variation in Switching Function S

Figure 7 illustrates the variation of the output voltage under the sliding mode control strategy. When the load current experiences a sudden change, a voltage oscillation of approximately 10V is observed. After the system reaches steady state, the output voltage under sliding mode control exhibits a continuous but very small incremental trend, eventually stabilizing at a constant level.

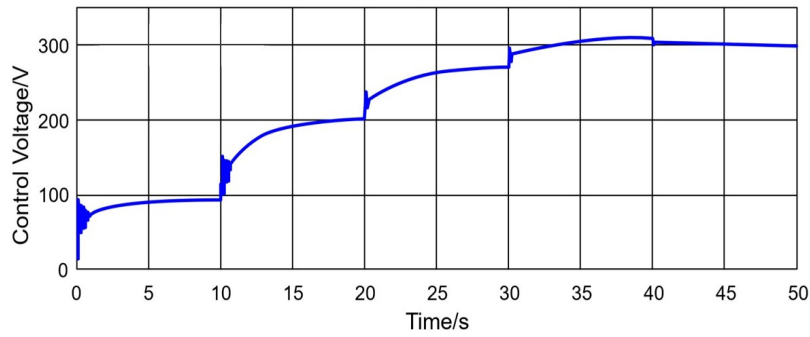


Figure 7. Output Voltage of SMC

Figure 8 illustrates the dynamic evolution of the oxygen excess ratio under sliding mode control. At 10s, the load current abruptly increases to 70A, leading to a sharp rise in oxygen consumption. As a result, the oxygen excess ratio rapidly decreases from its initial value of 2.31 to approximately 1.5. To meet the increased oxygen demand, the air compressor raises its output voltage, thereby increasing the inlet air flow rate. Consequently, the oxygen excess ratio gradually recovers and reaches a new equilibrium after approximately 7s. At 40 s, the load current decreases by 20A. During this transition, the air compressor continues supplying oxygen at the previous rate, causing a temporary increase in oxygen concentration. As the oxygen excess ratio rises, the compressor output voltage decreases accordingly, regulating the oxygen supply. Ultimately, the system automatically adjusts to a new reference level of the oxygen excess ratio, completing a smooth and stable transition.

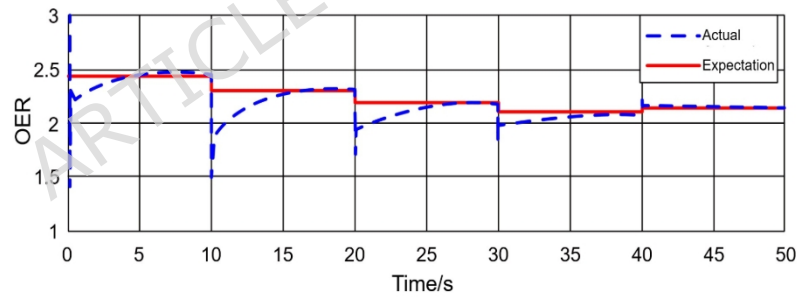


Figure 8. Oxygen Excess Ratio Under SMC

Figure 9 illustrates the variation of net power under the sliding mode control strategy. When a step disturbance occurs in the load current, the fluctuation amplitude of the system's net power output remains relatively small, indicating effective suppression of transient disturbances.

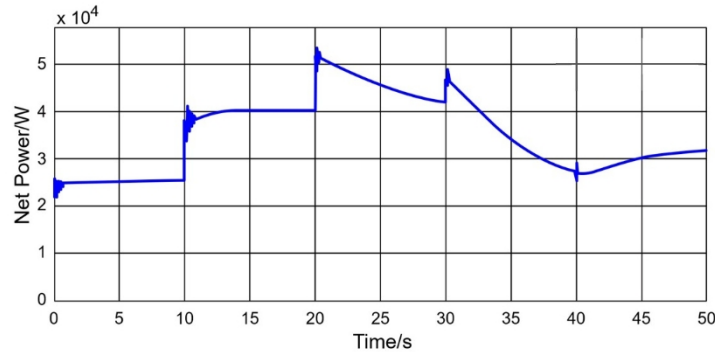


Figure 9. Net Power Under SMC

The advantages of the PEMFC gas supply system under sliding mode control are summarized as follows:

1. Strong robustness: Sliding mode control employs high-frequency switching control laws to constrain the system state on the sliding surface, demonstrating exceptional robustness against model parameter uncertainties (e.g., variations in flow coefficients or temperature) and external disturbances such as sudden load current changes.
2. Suitability for nonlinear systems: The hydrogen fuel cell gas supply system exhibits inherent nonlinear dynamics. Sliding mode control is directly designed based on this nonlinear model, thereby avoiding approximation errors that may arise from linearization.
3. Rapid dynamic response: The objective of the sliding mode control strategy is to drive the system state to the sliding surface within a finite time. Given the dynamic characteristics of fuel cell systems, particularly during abrupt load changes, immediate adjustment of the oxygen supply is essential to maintain stable system performance.

4.2 Analysis of Model Predictive Control Simulation Results for PEMFC Air Supply Subsystem

When designing the MPC-based DMC strategy, the MPC Toolbox is employed as the implementation platform, and the prediction horizon p is set to 40. This choice aims to balance system response speed and control accuracy, enabling the controller to effectively predict future system behavior and generate appropriate control actions. The selection of the prediction horizon considers system characteristics, operational constraints, and performance objectives to achieve optimal control performance. In this configuration, the control horizon M is required to be smaller than the prediction horizon. Accordingly, the control horizon M was set to 5. The controller output is constrained within the range of 1.8 to 2.5, while the maximum input voltage of the air compressor is limited to 300 V. Under these constraints, the parameters in the MPC Toolbox are appropriately tuned. After completing the controller configuration, the MPC Toolbox is applied to the linearized model. For the air supply system under step input conditions, calibrating the linearized model is a

critical step, as it enables more accurate prediction and control of the system's dynamic response.

It is worth noting that, although load variations within a certain range are involved here, the system's dynamic response remains relatively smooth within the operating conditions studied; therefore, a single-point linear model can still provide a reasonably good approximation of the system's behavior. For wider operating ranges or conditions with strong nonlinearity, gain scheduling or nonlinear MPC methods may be further employed.

Figure 10 illustrates the variation of the air compressor's output voltage under the MPC strategy. In this study, the air compressor is subject to a maximum input voltage constraint of 300 V. When the load current changes, the compressor voltage is automatically adjusted to regulate the oxygen excess ratio toward its reference value, reflecting the dynamic adaptability of the system. At $t = 13.56\text{s}$, the compressor voltage reaches the upper limit of 300 V and remains stable thereafter, indicating that the MPC controller is capable of effectively enforcing the input constraints. In contrast, the SMC strategy occasionally caused the compressor voltage to exceed 300 V. Although SMC can also effectively regulate the oxygen excess ratio, sustained high voltage may negatively impact the compressor, potentially reducing its service life.

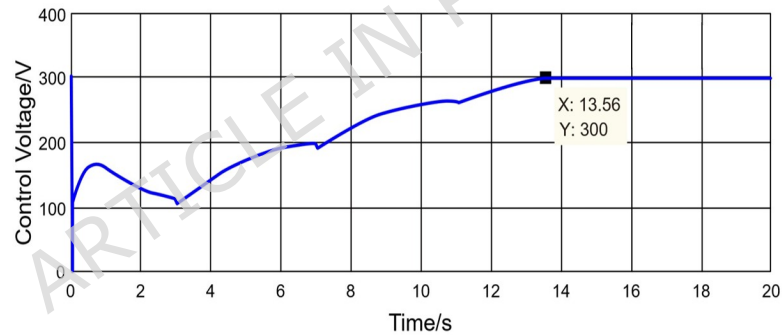


Figure 10. Controller Voltage Output for MPC

Figure 11 illustrates the dynamic evolution of the oxygen excess ratio under the MPC strategy. During the initial 2s, the oxygen excess ratio exhibits noticeable fluctuations, which are primarily attributed to the large deviation between the initial operating value and the desired reference. The MPC controller completes the regulation of the oxygen excess ratio within approximately 2s, indicating a relatively fast transient response. Although its response speed is slightly slower than that of sliding mode control, MPC demonstrates superior overall dynamic performance, making it more suitable for process-oriented control applications. Moreover, under steady-state conditions, the MPC strategy is able to maintain the oxygen excess ratio in close proximity to the reference value while respecting the voltage constraints of the air compressor. These results confirm the high reliability and effectiveness of the MPC-based control strategy.

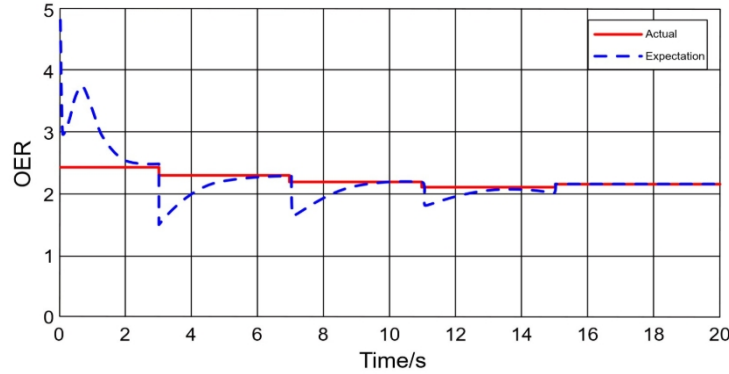


Figure 11: Oxygen Excess Ratio Curve Under MPC

Figure 12 presents the net power output of the proton exchange membrane fuel cell under the MPC strategy, which is noticeably higher than that obtained using the SMC approach. When the compressor voltage remains below 300V, the net power outputs under both control strategies are nearly identical, indicating comparable performance within this operating range. However, once the compressor voltage reaches the upper constraint of 300V, the MPC strategy exhibits a clear advantage by maintaining a significantly higher net power output. This improvement highlights the superior capability of MPC in handling actuator constraints while simultaneously optimizing system performance, in contrast to the sliding mode control strategy.

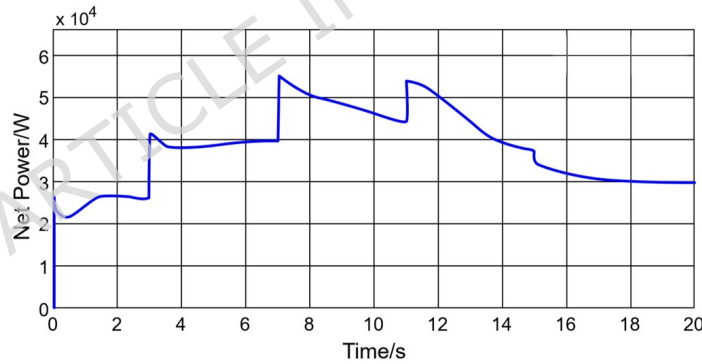


Figure 12. Net Power Curve Under MPC

The MPC controller designed in this paper achieves zero steady-state error in the oxygen excess ratio without inducing overshoot, thereby significantly improving the system's dynamic performance and overall energy efficiency compared with conventional sliding mode control. In particular, when addressing the voltage constraints of the air compressor, the proposed MPC strategy effectively avoids overvoltage risks while simultaneously maximizing the net power output of the system.

4.3 Simulation Verification Analysis of the PEMFC Air Supply Subsystem Under SMC-MPC Control

In this study, the test condition for the study involved selecting the load data obtained during the operation of the fuel cell system. At 5s, the load current increases stepwise from

20A to 35A and subsequently decreases to 25A at 12s. Figure 13 presents the load current response of the PEMFC system under three different control strategies. The results show that the SMC strategy provides a rapid dynamic response but suffers from noticeable oscillations and overshoot at the step transitions. In contrast, the MPC strategy yields smoother current variations; however, a certain response delay is observed. The proposed SMC-MPC hybrid control strategy ensures smooth variations in the control signal, thereby effectively avoiding abrupt fluctuations. Compared with the SMC strategy, it significantly suppresses oscillations, which helps mitigate the adverse effects associated with frequent actuator switching. Meanwhile, the control input consistently remains within a reasonable range, with no evident saturation observed.

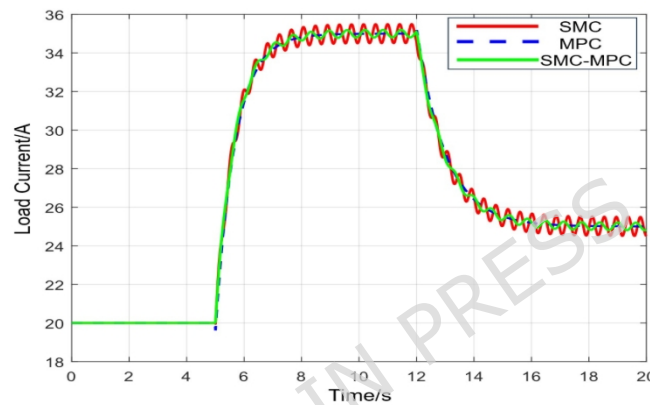


Figure 13. Comparison of Load Current in PEMFCs Under Three Control Strategies

Figure 14 illustrates the output voltage responses of the PEMFC system under three different control strategies. Voltage drops are observed in all cases during load current transients. The results show that the SMC strategy provides a rapid voltage response but is accompanied by noticeable fluctuations. In contrast, the MPC strategy produces a smoother voltage profile, albeit with a slight response delay. The proposed SMC-MPC hybrid control strategy effectively combines the advantages of both methods, achieving fast dynamic response with minimal and nearly negligible voltage fluctuations, while maintaining the output voltage within an optimal range throughout the entire load current variation process.

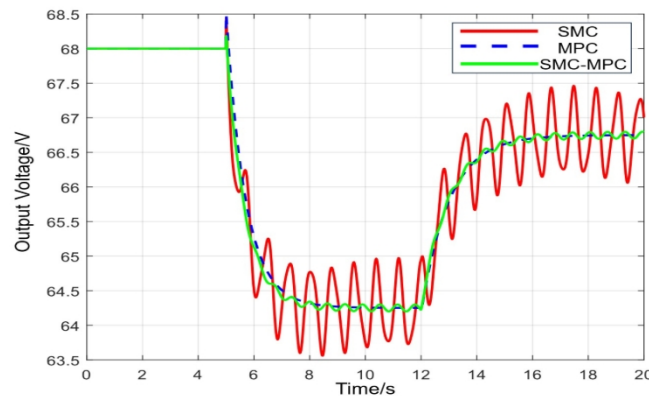


Figure 14. Comparison of Output Voltage for PEMFCs Under Three Control Strategies
 Figure 15 compares the net power output of the PEMFC system under three different control strategies. As the load varies, all three strategies dynamically adjust the system power output. The results indicate that the SMC strategy enables a rapid increase in net power but is accompanied by pronounced fluctuations. The MPC strategy exhibits a slower power rise and yields a comparatively lower peak power. In contrast, the proposed SMC-MPC hybrid control strategy achieves a fast power response with minimal fluctuations and delivers the highest steady-state net power among the three approaches.

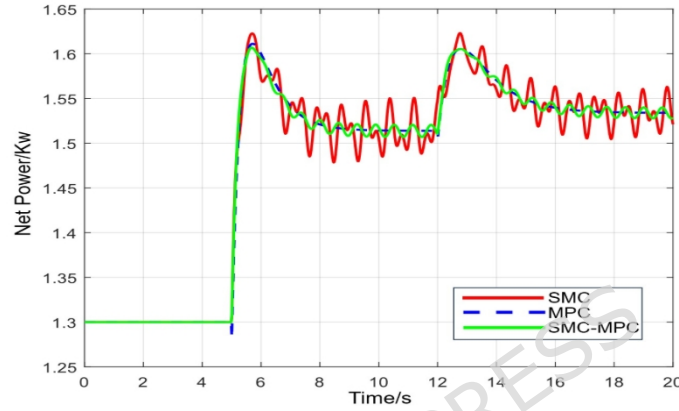


Figure 15. Comparison of Net Power Output of PEMFCs Under Three Control Strategies

Figure 16 illustrates the dynamic response of the oxygen excess ratio in the PEMFC system under three different control strategies. The results show that under SMC, the oxygen excess ratio exhibits pronounced fluctuations, mainly due to the inherent characteristics of conventional SMC, such as sensitivity to parameter variations and the chattering effect. In contrast, the MPC strategy provides a more stable response but with a slower dynamic adjustment, resulting in a noticeable deviation from the desired reference value. The proposed SMC-MPC hybrid control strategy enables rapid recovery of the oxygen excess ratio to a reasonable operating range, effectively suppressing large fluctuations while maintaining high gas utilization efficiency.

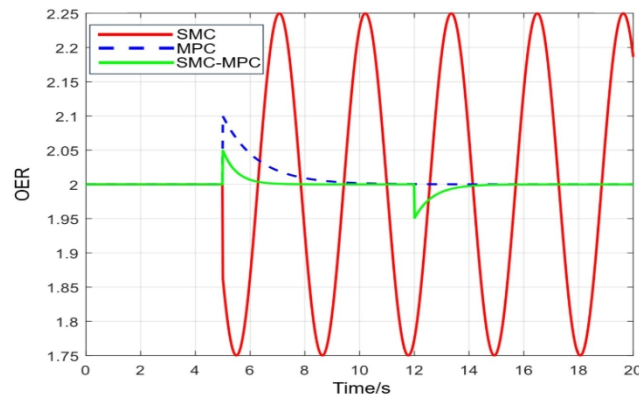


Figure 16. Comparison of Oxygen Excess Ratios in PEMFCs Under Three Control Strategies

Strategies

To comprehensively evaluate and compare the performance of the different control strategies, a quantitative analysis of the SMC, MPC, and SMC-MPC approaches was conducted using overshoot, settling time, steady-state error, and disturbance recovery time as the primary performance metrics. The key performance parameters of the three control methods are summarized in Table 3.

Table 3. Key Performance Parameter Comparison

Control Strategy	Overshoot	Adjustment time	Steady-state error	Disturbance recovery time
SMC	8.1%	1.5s	0.02	2.4s
MPC	4.5%	2.5s	0.01	3.0s
SMC-MPC	3.2%	1.2s	0.02	1.5s

(1) Overshoot: The SMC strategy exhibits a relatively large overshoot of 8.1%, indicating a fast response but at the expense of excessive transient deviation. In contrast, the MPC strategy achieves a lower overshoot of 4.5%; however, its slower dynamic response leads to a prolonged settling process. The proposed SMC-MPC hybrid control strategy effectively combines the advantages of both approaches, achieving the smallest overshoot of only 3.2% while maintaining a rapid response.

(2) Settling Time: SMC achieves a short settling time of 1.5s, reflecting its fast response capability, albeit accompanied by significant overshoot. MPC requires a longer settling time of 2.5s due to its reliance on rolling optimization to determine control inputs. The SMC-MPC hybrid controller demonstrates the best settling time performance, reducing the settling time to 1.2s while simultaneously avoiding excessive fluctuations.

(3) Steady-State Error: MPC exhibits the smallest steady-state error of 0.01, benefiting from its global optimization framework and accurate reference tracking capability. Although the steady-state error of the SMC-MPC hybrid control strategy is slightly larger, it remains well within acceptable limits for practical applications.

(4) Disturbance Recovery Time: The SMC-MPC hybrid controller demonstrates the fastest disturbance recovery performance, restoring system equilibrium within 1.5s, which highlights its strong robustness. In comparison, MPC requires 3.0s to recover from disturbances, while SMC achieves recovery in 2.4s.

In terms of implementation feasibility, the proposed SMC-MPC hybrid control strategy is capable of satisfying real-time control requirements. The MPC component adopts a finite prediction horizon and an appropriately selected sampling time, thereby effectively limiting the computational burden associated with online optimization. Meanwhile, the SMC component features a simple structure and low computational complexity, enabling rapid compensation without significantly increasing the overall computational load.

Overall, these results validate the superiority of the proposed SMC-MPC hybrid control strategy for PEMFC air supply systems. The hybrid approach not only achieves rapid dynamic response but also maintains low overshoot and fast disturbance recovery, while exhibiting excellent steady-state accuracy and disturbance suppression capability. Compared with standalone SMC or MPC strategies, the SMC-MPC hybrid controller provides a more balanced, robust, and efficient control solution.

5. Conclusion

This paper investigates advanced control strategies for PEMFC systems. Based on an innovative control framework that integrates global optimization with local compensation, a novel SMC-MPC hybrid control strategy is proposed. Comparative simulation studies of the SMC-MPC, SMC, and MPC strategies were conducted using a sliding mode control model of the PEMFC gas supply subsystem. The main conclusions can be summarized as follows:

1. Compared with standalone SMC and MPC strategies, the proposed SMC-MPC hybrid control strategy reduces overshoot by 60% and 28%, respectively; achieves the shortest settling time, with reductions of 20% and 52%; and provides the fastest disturbance recovery, shortening recovery time by 50% and 37.5%, respectively.

2. The SMC-MPC hybrid strategy inherits the strong robustness of SMC in resisting external disturbances and system uncertainties, while fully exploiting the advantages of MPC in multivariable optimization and constraint handling, thereby enabling effective global optimization under multiple operating conditions.

3. By addressing the inherent limitations of single control strategies in coping with the complex dynamics and disturbances of PEMFC systems, the proposed SMC-MPC approach significantly enhances dynamic response performance, system stability, and disturbance rejection capability.

6. Limitations and Perspective

Although the effectiveness of the proposed SMC-MPC hybrid control strategy has been verified through simulation, several limitations remain. First, the present study is primarily based on a simulation environment and does not fully account for practical factors such as sensor noise, actuator dynamics and delays, and model parameter uncertainties encountered in real systems. Second, the control parameters adopted in this study (e.g., the MPC prediction horizon and SMC gains) were determined under specific operating conditions, and a systematic sensitivity analysis of these parameters has not yet been performed. Future work will focus on validating the proposed strategy beyond simulation studies and further optimizing the control framework to address challenges such as real-time adaptive learning, scalability to large-scale systems, and fault detection and isolation, particularly under long-

term degradation and complex operating conditions. Moreover, the integration of digital twin technology with artificial intelligence is expected to provide intelligent real-time monitoring and optimization capabilities, thereby further improving the efficiency, durability, and sustainability of PEMFC systems.

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Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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